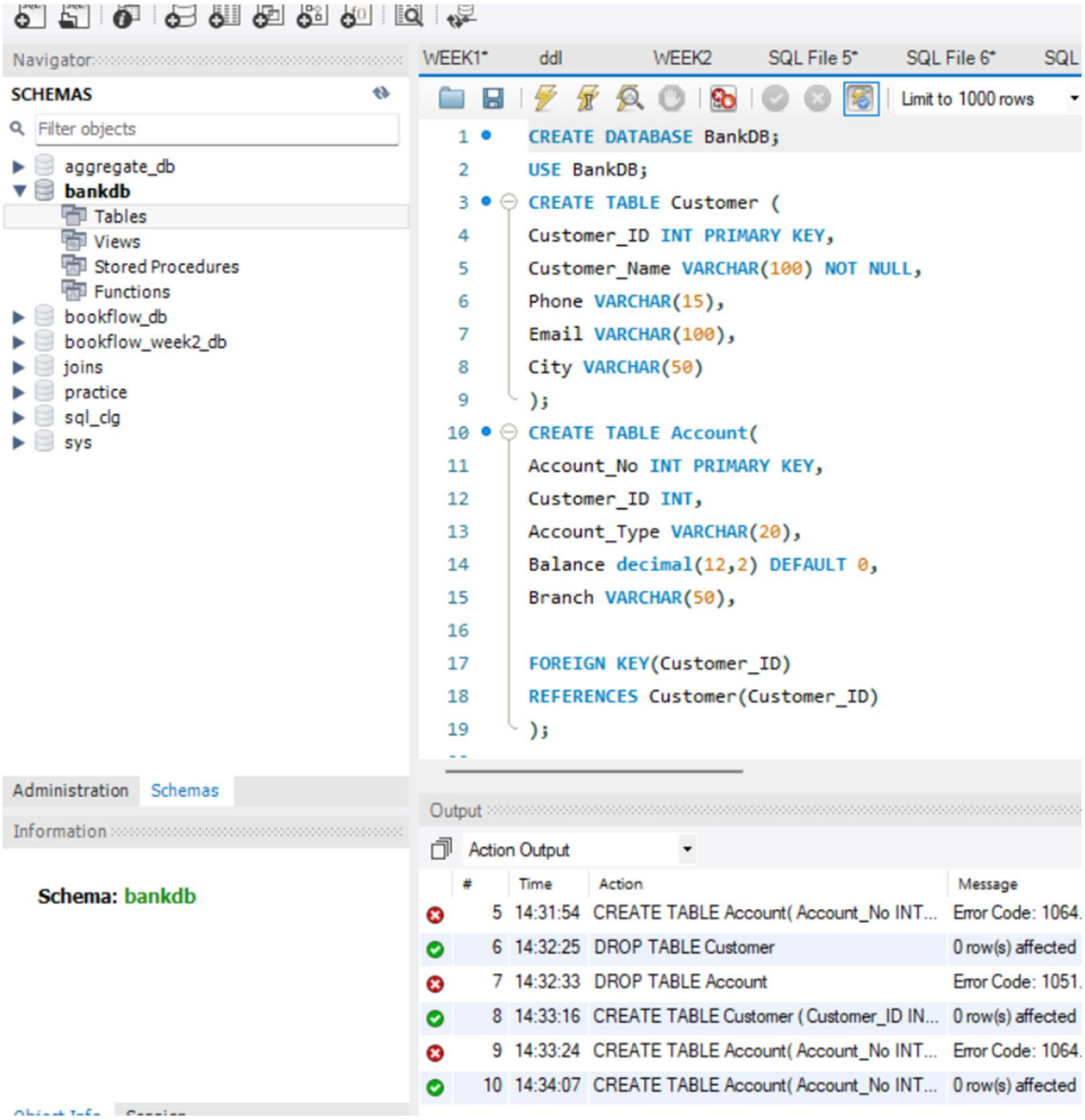
BANK DATABASE – TRIGGERS AND STORED PROCEDURES

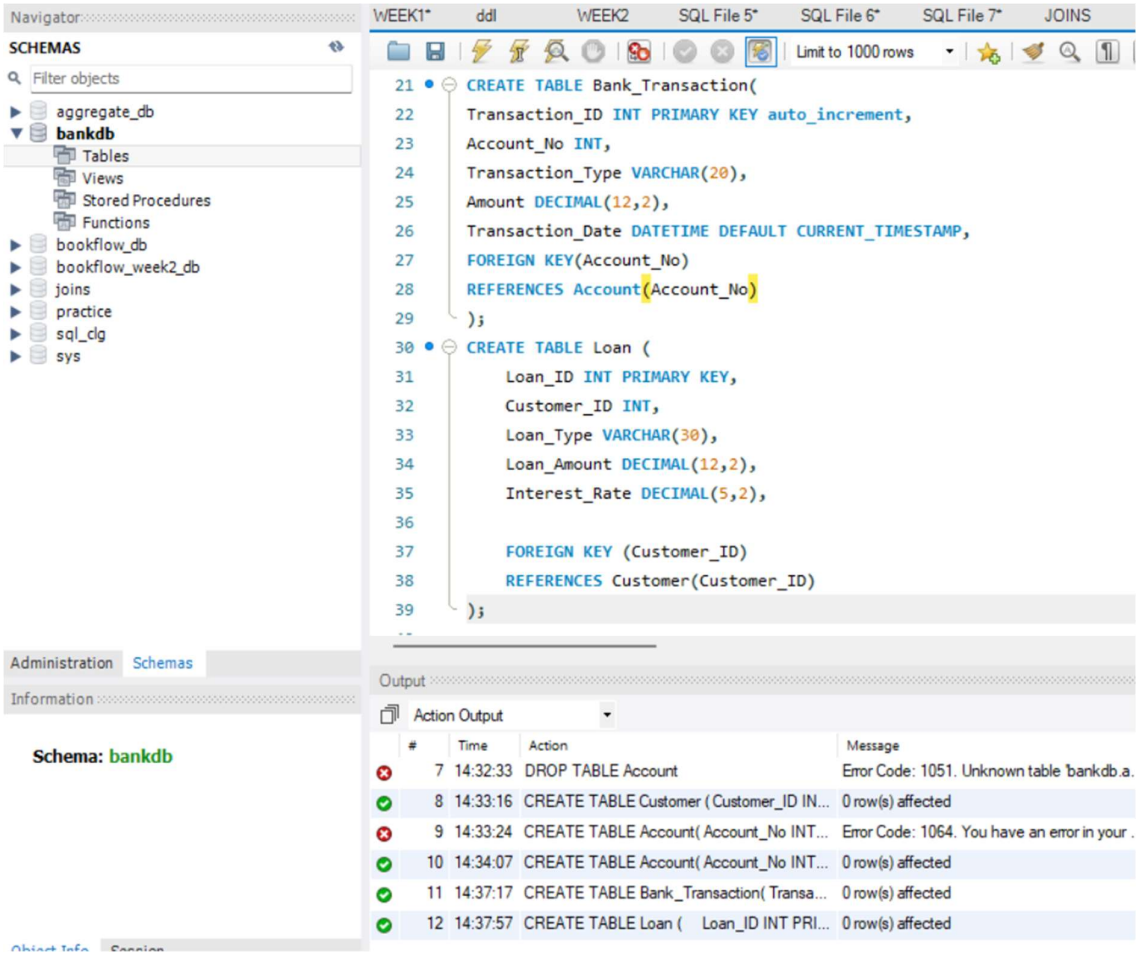
2520030071

G. Hemanth Venkatesh

1. CREATE DATABASE and Customer Table



2.CREATE TRANSACTION TABLE and LOAN TABLE



3.INSERT CUSTOMER DATA

```
39  );
40  INSERT INTO Customer
41  (Customer_ID, Customer_Name, Phone, Email, City)
42  VALUES
43  (101, 'Sai Akhil', '2520030021', 'saiakhil@gmail.com', 'Hyderabad'),
44  (102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),
45  (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
46  (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
47  (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
48
```

Output

Action Output

#	Time	Action	Message	Durati
8	14:33:16	CREATE TABLE Customer (Customer_ID IN...	0 row(s) affected	0.046 s
9	14:33:24	CREATE TABLE Account(Account_No INT...	Error Code: 1064. You have an error in your ...	0.000 s
10	14:34:07	CREATE TABLE Account(Account_No INT...	0 row(s) affected	0.031 s
11	14:37:17	CREATE TABLE Bank_Transaction(Transa...	0 row(s) affected	0.031 s
12	14:37:57	CREATE TABLE Loan (Loan_ID INT PRI...	0 row(s) affected	0.031 s
13	14:38:49	INSERT INTO Customer (Customer_ID, Cust...	5 row(s) affected Records: 5 Duplicates: 0 ...	0.015 s

4. INSERT ACCOUNT DATA

- joins
- practice
- sql_clg
- sys

```
45  (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
46  (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
47  (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
48
49  INSERT INTO Account
50  (Account_No, Customer_ID, Account_Type, Balance, Branch)
51  VALUES
52  (10001, 101, 'Savings', 50000, 'Hyderabad'),
53  (10002, 102, 'Savings', 75000, 'Vijayawada'),
54  (10003, 103, 'Current', 120000, 'Bangalore'),
55  (10004, 104, 'Savings', 45000, 'Chennai'),
56  (10005, 105, 'Current', 90000, 'Hyderabad');
```

dministration Schemas

nformation

Schema: bankdb

Output

Action Output

#	Time	Action	Message	Du
9	14:33:24	CREATE TABLE Account(Account_No INT...	Error Code: 1064. You have an error in your ...	0.0
10	14:34:07	CREATE TABLE Account(Account_No INT...	0 row(s) affected	0.0
11	14:37:17	CREATE TABLE Bank_Transaction(Transa...	0 row(s) affected	0.0
12	14:37:57	CREATE TABLE Loan (Loan_ID INT PRI...	0 row(s) affected	0.0
13	14:38:49	INSERT INTO Customer (Customer_ID, Cust...	5 row(s) affected Records: 5 Duplicates: 0 ...	0.0
14	14:40:01	INSERT INTO Account (Account_No, Custo...	5 row(s) affected Records: 5 Duplicates: 0 ...	0.0

5. INSERT TRANSACTION DATA

```
55      (10004, 104, 'Savings', 45000, 'Chennai'),
56      (10005, 105, 'Current', 90000, 'Hyderabad');
57
58      INSERT INTO Bank_Transaction
59      (Account_No, Transaction_Type, Amount)
60      VALUES
61      (10001, 'DEPOSIT', 10000),
62      (10002, 'DEPOSIT', 15000),
63      (10003, 'WITHDRAW', 20000),
64      (10004, 'DEPOSIT', 5000),
65      (10005, 'WITHDRAW', 10000);
```

Output

Action Output

#	Time	Action	Message
9	14:33:24	CREATE TABLE Account(Account_No INT...	Error Code: 1064. You have an
10	14:34:07	CREATE TABLE Account(Account_No INT...	0 row(s) affected
11	14:37:17	CREATE TABLE Bank_Transaction(Transa...	0 row(s) affected
12	14:37:57	CREATE TABLE Loan (Loan_ID INT PRI...	0 row(s) affected
13	14:38:49	INSERT INTO Customer (Customer_ID, Cust...	5 row(s) affected Records: 5 Du
14	14:40:01	INSERT INTO Account (Account_No, Custo...	5 row(s) affected Records: 5 Du

6.INSERT LOAN DATA

```
65      (10005, 'WITHDRAW', 10000);
66      INSERT INTO Loan
67      (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
68      VALUES
69      (501, 101, 'Home Loan', 5000000, 7.5),
70      (502, 102, 'Education Loan', 1000000, 6.5),
71      (503, 103, 'Car Loan', 800000, 8.2),
72      (504, 104, 'Personal Loan', 500000, 10.5);
```

Output

Action Output

#	Time	Action	Message
11	14:37:17	CREATE TABLE Bank_Transaction(Transa...	0 row(s) affected
12	14:37:57	CREATE TABLE Loan (Loan_ID INT PRI...	0 row(s) affected
13	14:38:49	INSERT INTO Customer (Customer_ID, Cust...	5 row(s) affected Records: 5 Duplicates: 0 ...
14	14:40:01	INSERT INTO Account (Account_No, Custo...	5 row(s) affected Records: 5 Duplicates: 0 ...
15	14:40:37	INSERT INTO Bank_Transaction (Account_...	5 row(s) affected Records: 5 Duplicates: 0 ...
16	14:40:55	INSERT INTO Loan (Loan_ID, Customer_ID...	4 row(s) affected Records: 4 Duplicates: 0 ...

7. CHECK EVERYTHING

```
73 •  SELECT * FROM Customer;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

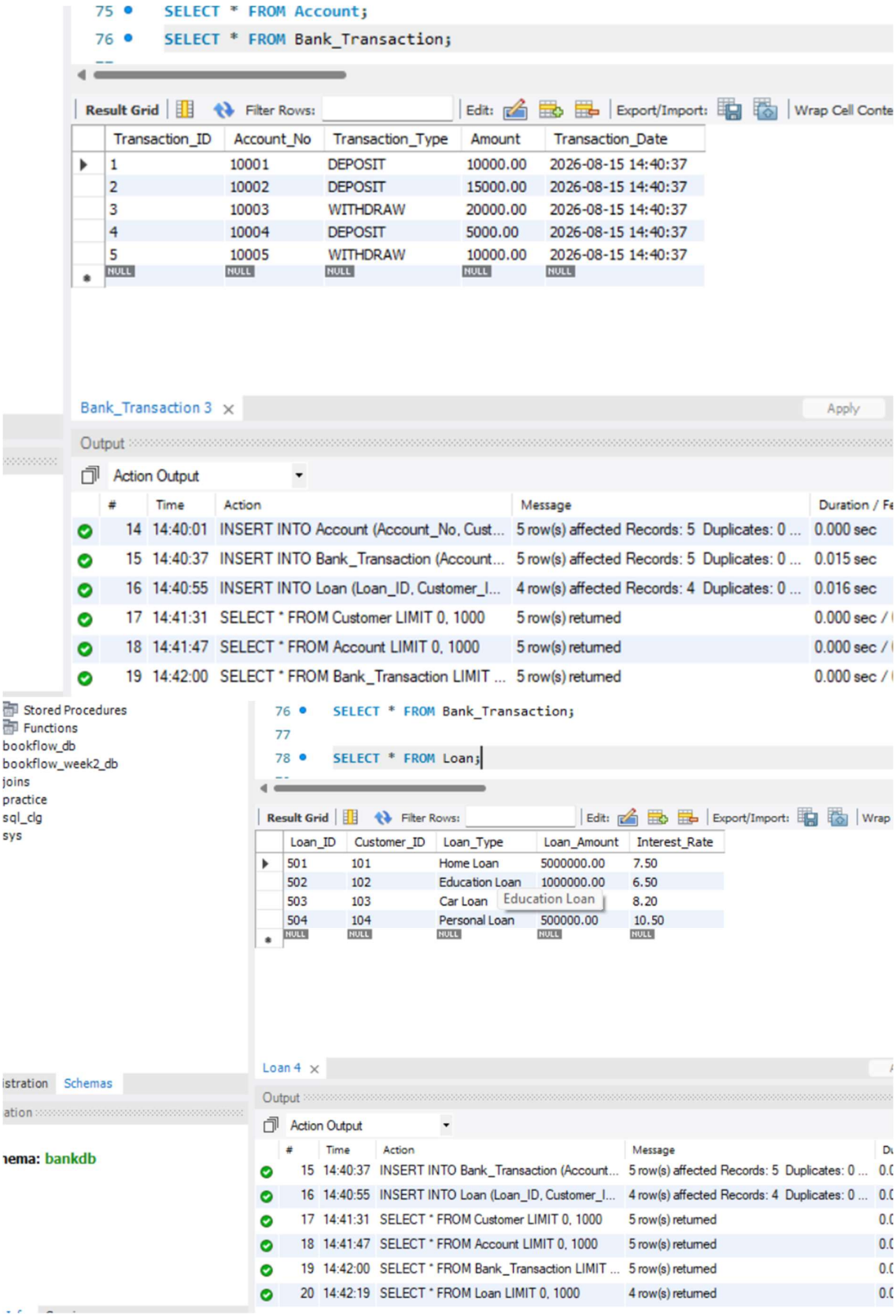
Customer_ID	Customer_Name	Phone	Email	City
101	Sai Akhil	2520030021	saiakhil@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
NULL	NULL	NULL	NULL	NULL

Customer 1 ×

```
74
75 •  SELECT * FROM Account;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL



8. PROCEDURE TO DISPLAY ALL CUSTOMERS

```
81 •  CREATE PROCEDURE GetAllCustomers()
82    BEGIN
83        SELECT * FROM Customer;
84    END //
85
86    DELIMITER ;
87 •  CALL GetAllCustomers();
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Phone	Email	City
101	Sai Akhil	2520030021	saiakhil@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

Result 5

Output

Action Output

#	Time	Action	Message	Durat
17	14:41:31	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000
18	14:41:47	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000
19	14:42:00	SELECT * FROM Bank_Transaction LIMIT ...	5 row(s) returned	0.000
20	14:42:19	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned	0.000
21	14:43:36	CREATE PROCEDURE GetAllCustomers() ...	0 row(s) affected	0.031
22	14:44:14	CALL GetAllCustomers()	5 row(s) returned	0.000

9. PROCEDURE TO DISPLAY ACCOUNT DETAILS

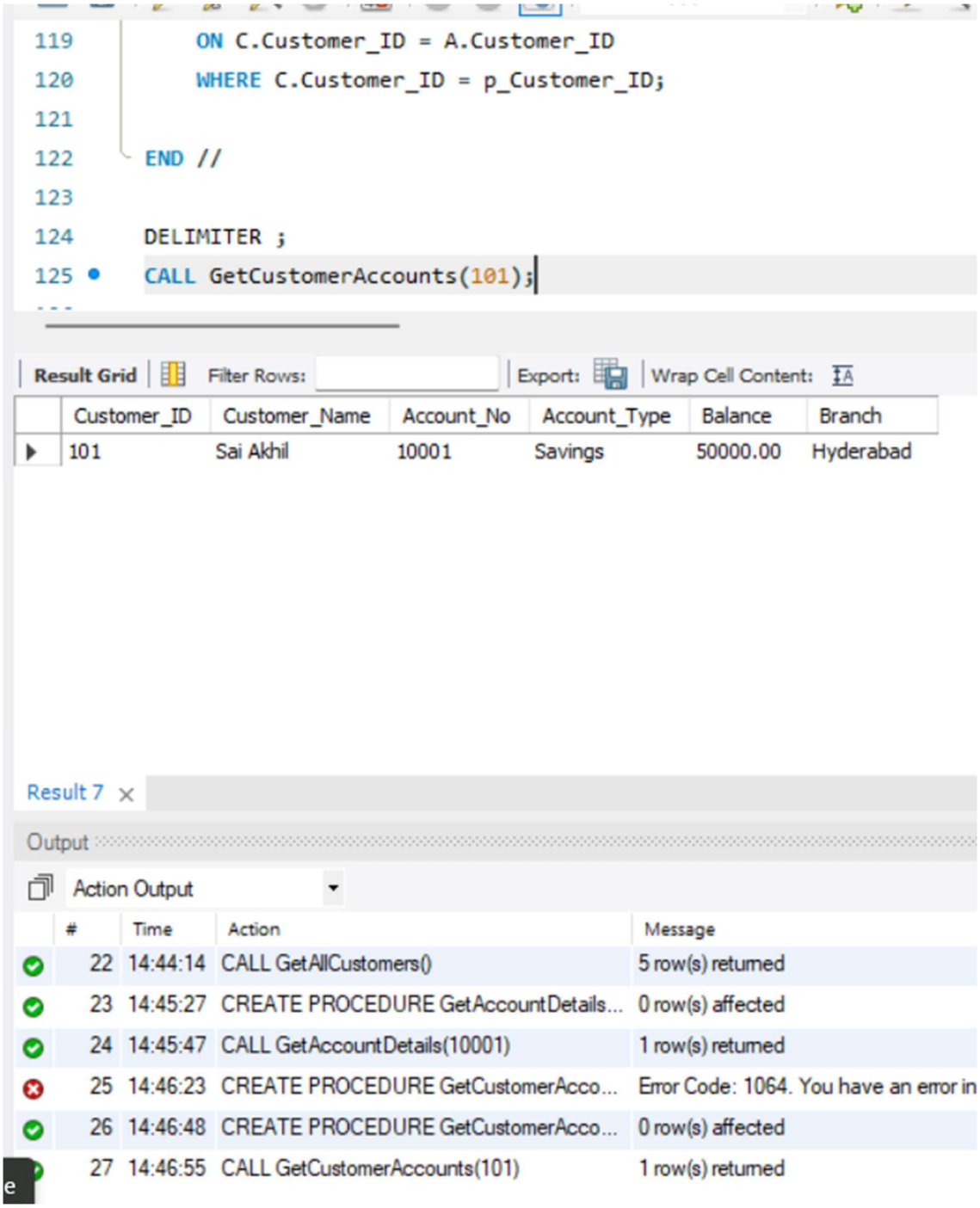
FOR A GIVEN ACCOUNT NUMBER

```
96      FROM Account
97      WHERE Account_No = p_Account_No;
98
99  END //
.00
.01  DELIMITER ;
.02  CALL GetAccountDetails(10001);
```

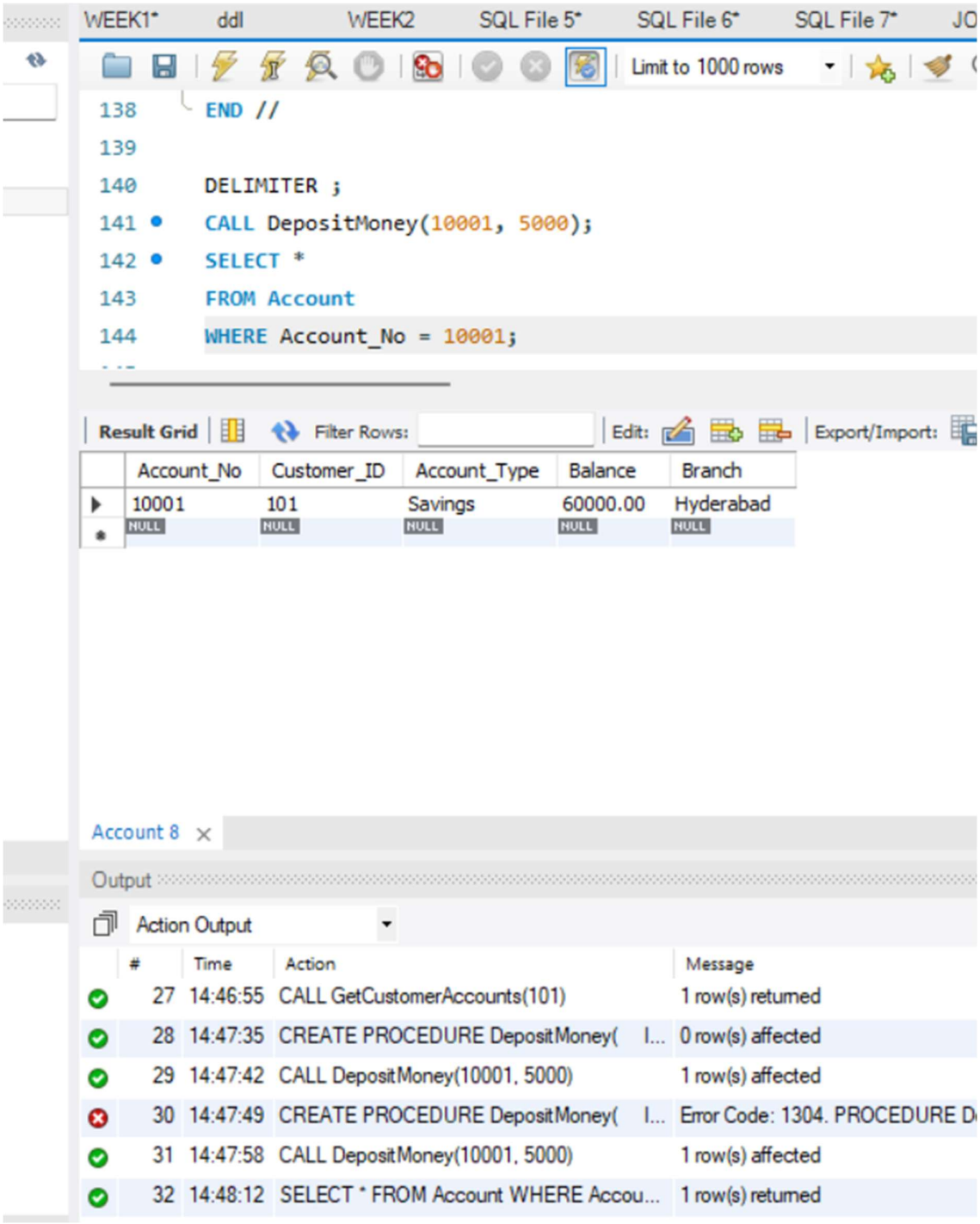
esult Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

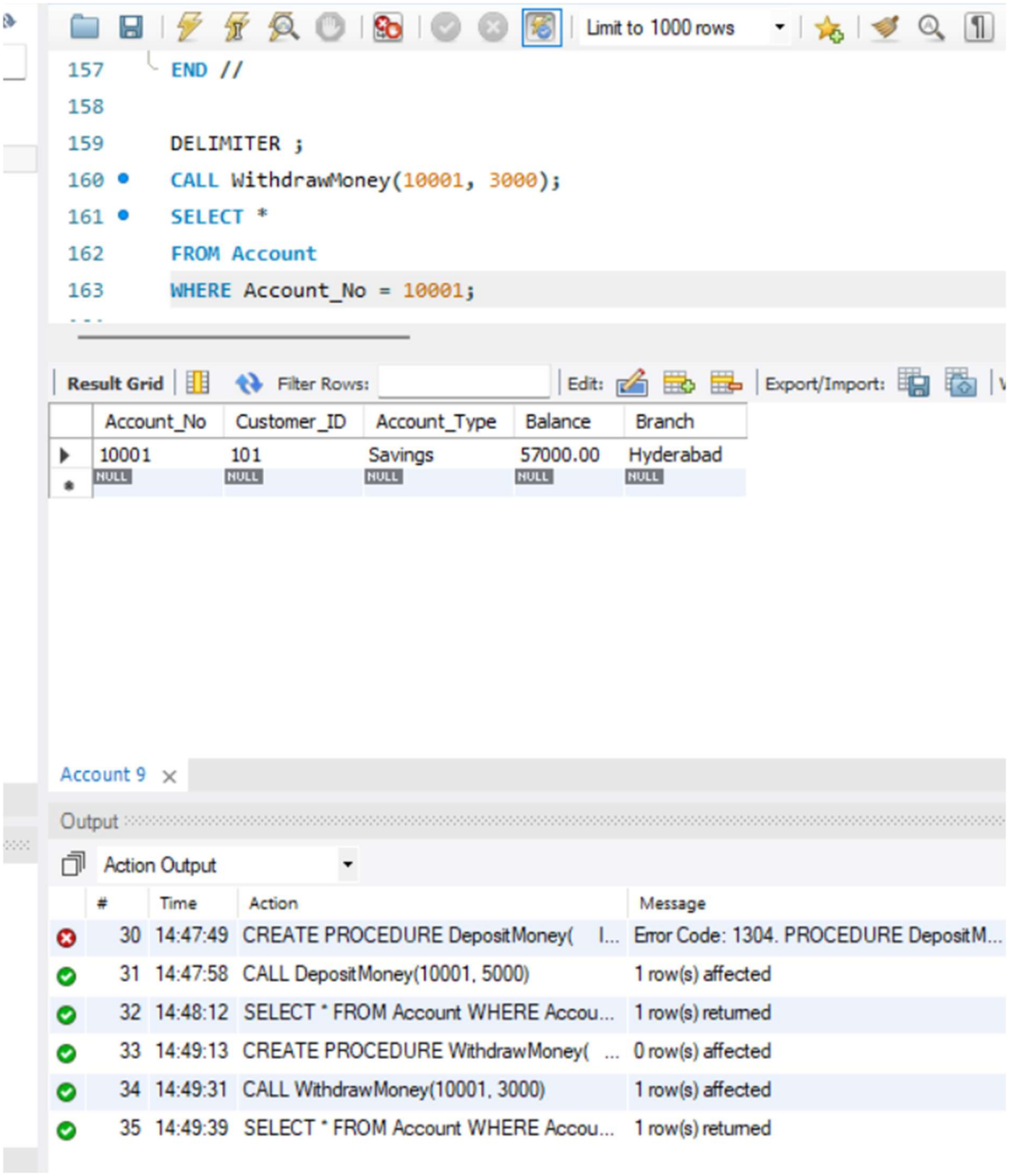
10. PROCEDURE TO FIND CUSTOMER ACCOUNTS



11. PROCEDURE TO DEPOSIT MONEY



12. Procedure — Withdraw Money



13.Trigger — Prevent Negative Balance

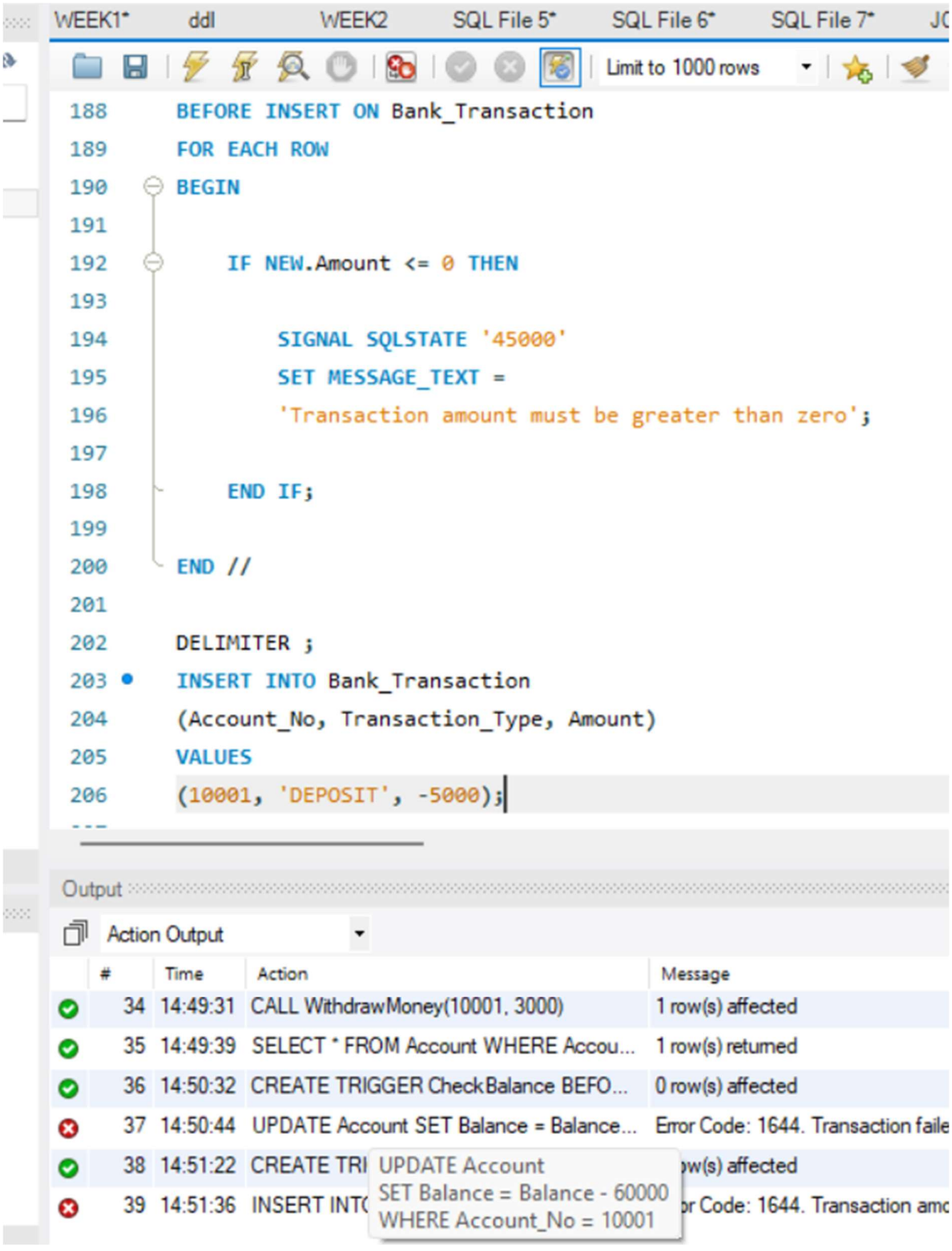
```
166 •  CREATE TRIGGER CheckBalance
167    BEFORE UPDATE ON Account
168    FOR EACH ROW
169    BEGIN
170
171        IF NEW.Balance < 0 THEN
172
173            SIGNAL SQLSTATE '45000'
174            SET MESSAGE_TEXT =
175            'Transaction failed: Insufficient balance';
176
177        END IF;
178
179    END //
180
181    DELIMITER ;
182 •  UPDATE Account
183    SET Balance = Balance - 60000
184    WHERE Account_No = 10001;
```

Output

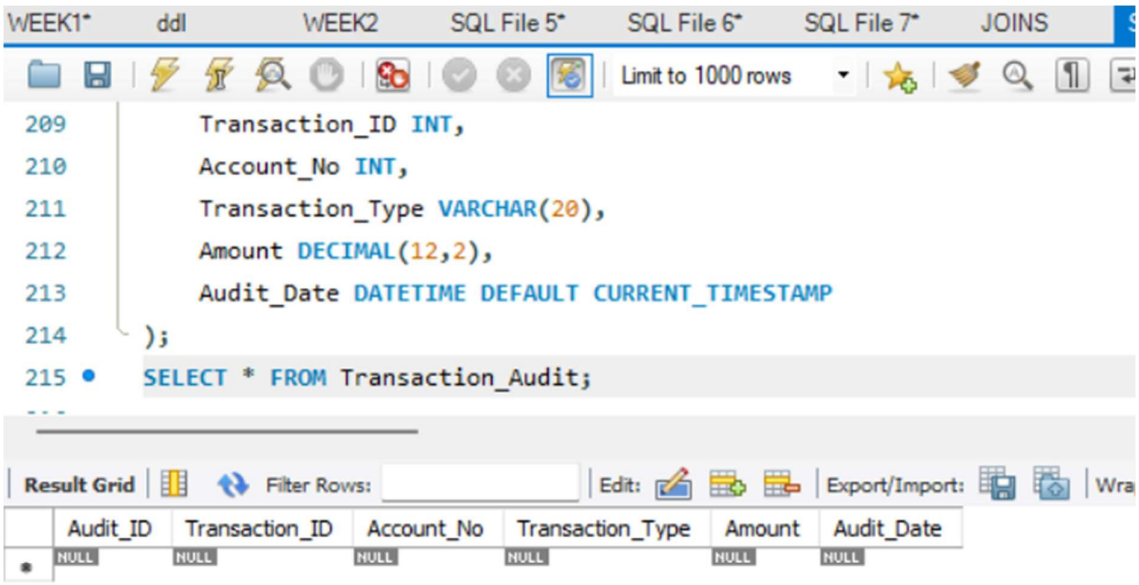
Action Output

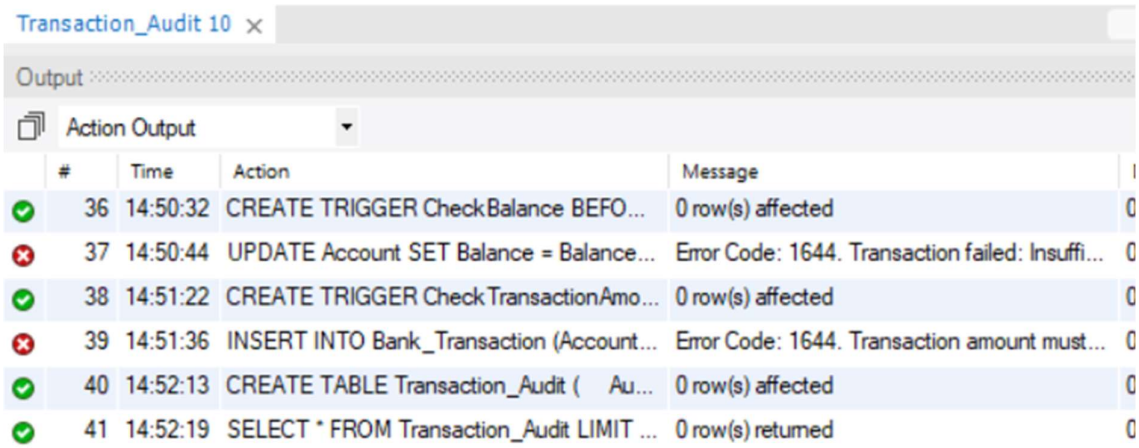
#	Time	Action	Message	Du
32	14:48:12	SELECT * FROM Account WHERE Accou...	1 row(s) returned	0.0
33	14:49:13	CREATE PROCEDURE WithdrawMoney(...	0 row(s) affected	0.0
34	14:49:31	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.0
35	14:49:39	SELECT * FROM Account WHERE Accou...	1 row(s) returned	0.0
36	14:50:32	CREATE TRIGGER CheckBalance BEFO...	0 row(s) affected	0.0
37	14:50:44	UPDATE Account SET Balance = Balance...	Error Code: 1644. Transaction failed: Insuffi...	0.0

14. Trigger — Prevent Negative Transaction Amount

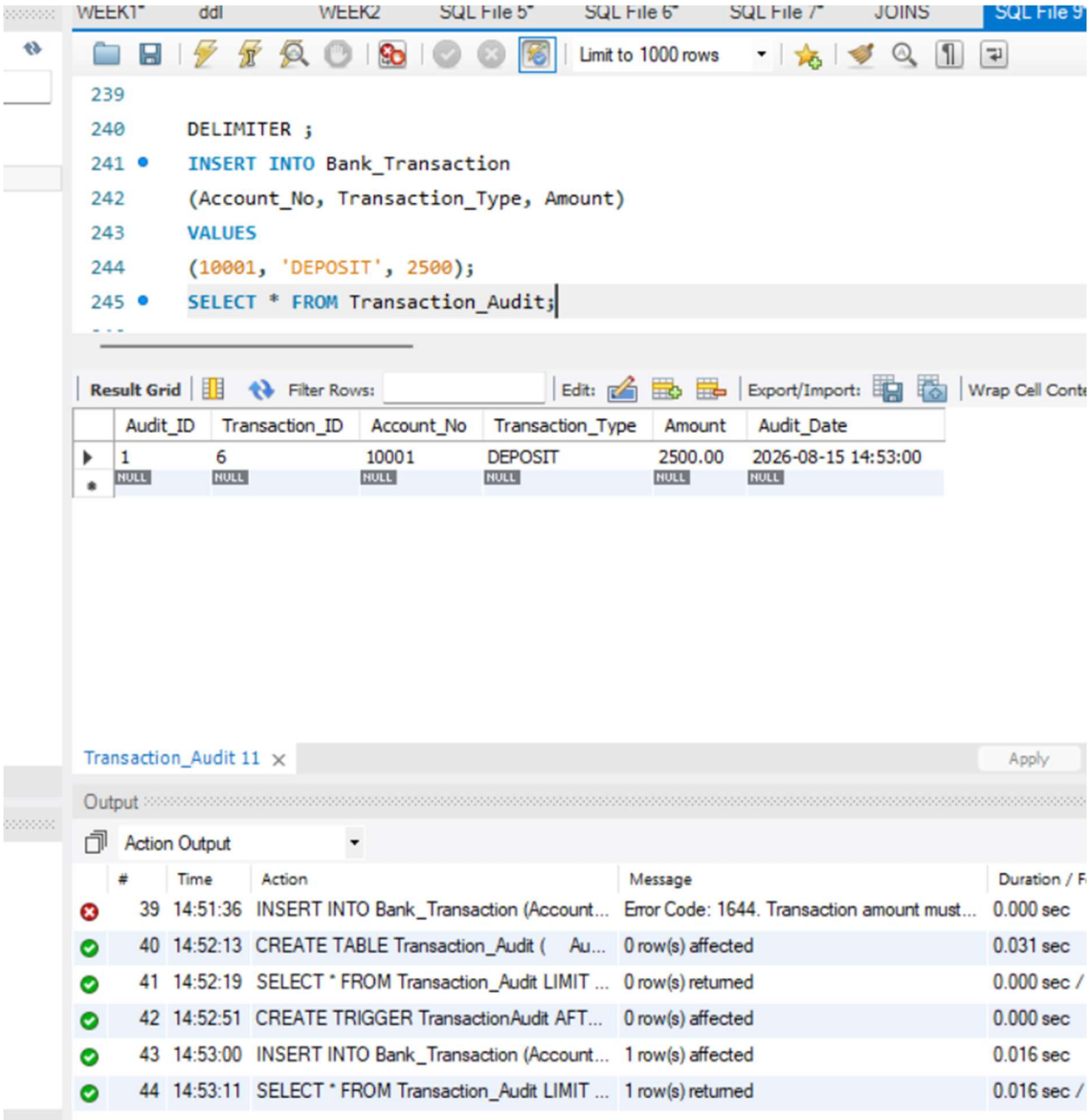


15. Create Transaction Audit Table





16. Trigger — Transaction Audit


WEEK1*
ddl
WEEK2
SQL File 5*
SQL File 6*
SQL File 7*
JOINS
SQL File 9*
Limit to 1000 rows
239
240 DELIMITER ;
241 INSERT INTO Bank_Transaction
242 (Account_No, Transaction_Type, Amount)
243 VALUES
244 (10001, 'DEPOSIT', 2500);
245 SELECT * FROM Transaction_Audit;
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Conte
Audit_ID | Transaction_ID | Account_No | Transaction_Type | Amount | Audit_Date
1 | 6 | 10001 | DEPOSIT | 2500.00 | 2026-08-15 14:53:00
NULL | NULL | NULL | NULL | NULL | NULL
Transaction_Audit 11
Apply
Output
Action Output
| Time | Action | Message | Duration / F
39 | 14:51:36 | INSERT INTO Bank_Transaction (Account... | Error Code: 1644. Transaction amount must... | 0.000 sec
40 | 14:52:13 | CREATE TABLE Transaction_Audit (Au... | 0 row(s) affected | 0.031 sec
41 | 14:52:19 | SELECT * FROM Transaction_Audit LIMIT ... | 0 row(s) returned | 0.000 sec /
42 | 14:52:51 | CREATE TRIGGER TransactionAudit AFT... | 0 row(s) affected | 0.000 sec
43 | 14:53:00 | INSERT INTO Bank_Transaction (Account... | 1 row(s) affected | 0.016 sec
44 | 14:53:11 | SELECT * FROM Transaction_Audit LIMIT ... | 1 row(s) returned | 0.016 sec /